

NK Annotation Biomarkers and Agent Logic

Current annotation-agent reference after removing broad curated non-NK panels and switching taxonomy DE evidence to balanced top positive/top negative genes.

Sources: marker_knowledge.py, taxonomy_reference.py, evidence.py, prompts.py, graph.py, and FINAL_UNIFIED_NK_TAXONOMY_REFERENCE_nolayer.md.

1. Core Pan-NK Biomarkers

These are explicitly summarized to help distinguish NK-supported clusters from clusters where NK identity is weak. In cluster-vs-rest DE, low pan-NK logFC means lower than other mostly-NK clusters, not necessarily absent NK identity.

NCR1, KLRD1, KLRF1, NCAM1, EOMES, TBX21, NKG7, GNLY, PRF1, GZMB, GZMA, CST7

2. Curated Non-NK Sanity Markers

Only T-lineage markers are kept as a curated non-NK sanity panel because T and NK programs overlap and are a common annotation boundary. Broad B-cell, myeloid, erythroid, stromal, and epithelial marker panels are not provided as curated exclusion panels; those identities are inferred only if their genes appear naturally in the cluster DE evidence.

Panel	Markers
T	CD3D, CD3E, CD3G, TRAC, TRBC1, TRBC2

3. Marker Program Hints

Marker-program hints are lightweight summaries of the cluster top DE genes. They now focus on NK biology, proliferation/cytokine programs, and the T/NK boundary; they do not include broad curated B/myeloid/erythroid/stromal panels.

Program	Markers
NK cytotoxic	NKG7, GNLY, PRF1, GZMB, GZMH, GZMA, CST7, FGFBP2, KLRF1, FCGR3A
NK tissue/regulatory/chemokine	XCL1, XCL2, KLRC1, KLRC2, KLRB1, CXCR6, ITGAE, ZNF683, CCL3, CCL4, CCL5
proliferation	MKI67, TOP2A, STMN1, TYMS, RRM2, TK1, PCNA, PCLAF, NUSAP1
interferon/cytokine	ISG15, IFIT1, IFIT2, IFIT3, IFI44L, MX1, STAT1, IRF7, IL2RA, IL7R, CCR7, IRF4
T cell	CD3D, CD3E, CD3G, TRAC, TRBC1, TRBC2, TCF7, SELL, LEF1

4. Amina NK Taxonomy Biomarker Reference

The Amina taxonomy reference currently contributes 17 subtype/state marker entries. Each entry has core, support, context, and expected-low/negative markers.

Layer	Label	Core markers	Support markers	Context markers	Expected low / negative
subtype	NK2	GZMK, IL2RB, SELL, XCL1, XCL2, CCR7, NCAM1, IL7R, KLRC1, NCR1	CXCR3, SPINK2, AREG, GPR183, IKZF2, KIT, KLRK1, LTβ, NCR2, TCF7, IL18R1, IL18RAP, PTGDS, FLT3LG	EEF1A1, TNFRSF18, TPT1, S1PR1	
subtype	NK1	CST7, CX3CR1, FCGR3A, FGFBP2, GZMB, NKG7, PRF1, SPON2, GNLY, KLRD1	ADGRG1, GZMH, GZMM, S1PR5, ZEB2, ACTB, FCER1G, HAVCR2, KLRG1, PRDM1, TBX21, CD160, CHST2, CLIC3, KLF2, LAIR2, SIGLEC7, CFL1, CTSW, GZMA, RAC2		
subtype	adaptive_NK_CMV	KLRC2, B3GAT1, LILRB1, PRDM1, IL32, ZBTB38, FCGR3A	GZMH, CCL5, CX3CR1, TBX21, ZEB2, KLRD1, FGFBP2, SPON2, IFNG, CD244, CD2, FCGR2C, ITGAM	ARID5B, JAKMIP1, PATL2, KIR2DL1, KIR3DL1	FCER1G, SYK, ZBTB16, KLRC1, KLRB1, CCR7, IL2RB, TYROBP, SIGLEC7
subtype	adaptive_NK_nonCMV	PRDM1, IL32, BATF, MAF, CCL5, FCGR3A	GZMH, CD52, ZBTB32, ZBTB38, KIR2DL2, KIR2DL1, KIR3DL1, LILRB1, TMSB4X	CD3E, CX3CR1, TBX21, KLRC2	
subtype	trNK	CD69, EOMES, CXCR6, ITGA1	TBX21, AREG, ITGB1, CCL4L2, HOPX, IKZF3, ITGAE, RGS1, ZNF683	PDCD1, PSMA2, SCGB1A1, SLC5A3, CLN3	
subtype	cNK	CX3CR1	FCGR3A, S1PR5, KLRG1, KLF2, B3GAT1	IKZF1	
subtype	L6_Developmental_immature		NCAM1, SPINK2, GATA3, KIT, TCF7, ID2, MYC, CD34, FLT3, LEF1, RUNX2	IL2RA, IL3RA	
state	Chemokine_inflammatory	XCL1, XCL2, CCL4, CCL5, CCL3, IFNG	STAT1, CCL4L2, GZMA, CXCL10, CXCL9		
state	Checkpoint_exhausted	TIGIT, HAVCR2, TOX, NR4A2, NR4A1, LAG3, PDCD1, ENTPD1	KLRC1, CD96, TOX2, BATF, LILRB1, SIGLEC7, KIR2DL1, KIR3DL1, NR4A3, CTLA4, CD274, LAYN, CISH, SOCS1, SOCS3, NFATC1, NFATC2, BHLHE40, IRF4, EZH2, CD38, NT5E, RELB, NFKB2, RXRA, TNFRSF9	PRDM1, CD244, CD160, KLRG1, CX3CR1, EOMES, IKZF2, ZEB2, MAF, MAFB, EGR2, EGR3, RGS1, BATF3, CDKN1A, HIF1A	
state	ER_stress_UPR		CD44, CD74, CXCR4, MAFF, ATF3, DDIT3, DNAJB1, EGR2, HSPA1A, HSPA1B, XBP1, EGR3	DUSP1, GADD45B, BAG3	
state	Metabolic_stress_hypoxia		LDHA, MYC, NFE2L2, TXNIP, BNIP3, HIF1A, HK2, MTOR, PDK1, PFKF, SLC2A1	PRDX1, GCLC, SOD2	
state	Proliferating	MKI67	AURKB, CCNB1, CDK1, PCNA, TOP2A, TYMS, UBE2C, BIRC5, CCNB2, STMN1		
state	IFN_stimulated	ISG15, MX1, MX2	BST2, IFI44L, IFIT1, IRF7, OAS1, STAT1, IFIT2, IFIT3	RSAD2, STAT2, HERC5, OAS3, DDX58, OAS2	
state	Cytotoxic_activated	FCGR3A, GZMB, NKG7, PRF1, GNLY	CD160, KLRK1, NCR1, CTSW, GZMA, NCR3		
state	Homeostatic_quiescent	CST7	GNLY, KLRD1, CD244, CHST12, IL18RAP, KLRF1		
state	CIML_cytokine_preactivated		GZMB, IL2RB, PRF1, KLRC2, IL12RB2, IFNG, IL12RB1, STAT4	CD44, CD122, IL15RA	
state	CIMP_cytokine_primed_memory_like	IL2RA, IFNG, IL12RB1, IL12RB2, STAT4, PRF1, GZMB, GZMK, NCR2, NCR3, TCF7	TNFRSF9, RUNX3, ID2, CD44, NFIL3, JAK2, STAT5A, STAT5B, TNFSF14, LTβ, LTA, CSF2, IFNGR2, BCL2, ZBTB32, IRF8, TNF, CD96	OSM, LIF	CXCR4, S1PR5, NR4A1, NR4A2, NR4A3

5. Taxonomy DE Selection and Scoring

- Positive taxonomy DE candidates: adjusted $P < 0.05$, $\log_{2}FC > 0.25$, and pct in cluster ≥ 0.1 .
- Negative taxonomy DE candidates: adjusted $P < 0.05$, $\log_{2}FC < -0.25$, and pct in reference ≥ 0.1 .
- The code keeps up to 50 positive genes and up to 50 negative genes per cluster, for at most 100 taxonomy DE genes total.
- Positive genes are ranked by $\log_{2}FC$ descending, then adjusted P ascending. Negative genes are ranked by $\log_{2}FC$ ascending, then adjusted P ascending.
- Taxonomy marker score is a precomputed evidence summary: CORE hits = 4, SUPPORT hits = 2, CONTEXT hits = 1. `percent_of_max_score` is this weighted score divided by the maximum possible score for that taxonomy entry.
- The score is not the final label. It ranks taxonomy marker matches shown to the LLM; the final label is chosen from taxonomy hits, top DE genes, pan-NK evidence, T-lineage sanity evidence, metadata context, and pairwise DE when available.

Support-level thresholds

- strong: at least 3 CORE hits, or at least 2 CORE + 3 SUPPORT hits, or at least 1 CORE + 5 SUPPORT hits.
- moderate: at least 2 CORE hits, or at least 1 CORE + 2 SUPPORT hits, or at least 4 SUPPORT hits.
- weak: any smaller positive marker-hit pattern, or at least 2 expected-low markers depleted.
- contradictory: negative-defining markers are positive, especially 2 or more contradictions, or 1 contradiction with fewer than 2 positive hits.

6. Agent Annotation Logic

- For each Leiden cluster, the agent sees top positive and negative cluster-vs-rest DE genes, balanced taxonomy DE evidence, Amina taxonomy marker hits, pan-NK marker summaries, T-lineage sanity summaries, marker program hints, metadata context, and pairwise DE evidence when available.
- The agent chooses an allowed NK subtype and state, or Non-NK. `final_structured_label` is training-ready: `<subtype>_<state>`, or Non-NK.
- `free_label` is a concise biological interpretation that can include context such as proliferation, stress, tissue context, contamination concern, or key marker support.
- Weak pan-NK DE alone should not force Non-NK because the background is mostly NK/NK-enriched.
- For T/NK ambiguity, the curated T-lineage sanity check is used together with pan-NK support. Shared NK/T activation, cytotoxicity, chemokine, cytokine, or stress genes are not enough by themselves to call T/Non-NK.
- For B, myeloid, erythroid, stromal, epithelial, or other non-NK identities, the agent relies on the actual top DE genes in that cluster, not on curated exclusion panels.
- Pairwise DE is mainly used after a structured label is chosen, to decide whether clusters with the same structured label deserve different free labels.
- The graph normalizes LLM output to allowed labels and applies safety fallbacks when labels are missing or inconsistent. Human-review flags capture mixed evidence, contamination/stress concerns, or weak confidence.

7. Practical Interpretation Notes

- Cluster-vs-rest DE is relative to the analysis background. In an NK-enriched object, a marker can be negative relative to other NK clusters while still being expressed biologically.
- A Non-NK call should be based on coherent lineage evidence and weak/depleted NK support, not on a single marker or generic stress/proliferation signal.
- The structured label is meant for downstream model training; the free label and review flag preserve nuance for human interpretation.